PADHAKU
Mathematics · Class 10 Mathematics



CHAPTER 11

Areas Related to Circles

NCERT Exercise Solutions

Exercise 11.1	14 Q

✨ Try each question yourself first, then check the answer. 💡 answers are samples: write your own.

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Exercise 11.1

1 **Find the area of a sector of a circle with radius 6 cm if angle of the sector is 60°.**

✔ ANSWER

Given: r = 6 cm, θ = 60°

Area of sector = (θ/360°) × πr²

= (60/360) × (22/7) × 6 × 6

= (1/6) × (22/7) × 36 = (22 × 6)/7

Area = 132/7 cm² (≈ 18.86 cm²)

2 **Find the area of a quadrant of a circle whose circumference is 22 cm.**

✔ ANSWER

Circumference 2πr = 22

2 × (22/7) × r = 22 ⇒ r = 7/2 cm

A quadrant has angle 90°, so area = (90/360) × πr² = ¼ πr²

= ¼ × (22/7) × (7/2) × (7/2) = ¼ × (22 × 7)/4 = 154/16

Area of quadrant = 77/8 cm² (= 9.625 cm²)

3 **The length of the minute hand of a clock is 14 cm. Find the area swept by the minute hand in 5 minutes.**

✔ ANSWER

Minute hand turns 360° in 60 minutes, so in 5 minutes it turns (360/60) × 5 = 30°.

Area swept = sector of angle 30°, r = 14 cm

= (30/360) × (22/7) × 14 × 14

= (1/12) × 22 × 28 = 616/12

Area swept = 154/3 cm² (≈ 51.33 cm²)

4 **A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the corresponding: (i) minor segment (ii) major sector. (Use π = 3.14)**

✔ ANSWER

r = 10 cm, θ = 90°

Area of minor sector = (90/360) × 3.14 × 10 × 10 = 78.5 cm²

Area of ΔOAB (right angle at O) = ½ × OA × OB = ½ × 10 × 10 = 50 cm²

(i) Area of minor segment = sector − triangle = 78.5 − 50 = **28.5 cm²**

(ii) Area of circle = 3.14 × 10² = 314 cm²

Area of major sector = 314 − 78.5 = **235.5 cm²**

5 **In a circle of radius 21 cm, an arc subtends an angle of 60° at the centre. Find: (i) the length of the arc (ii) area of the sector formed by the arc (iii) area of the segment formed by the corresponding chord.**

✔ ANSWER

r = 21 cm, θ = 60°

(i) Length of arc = (θ/360) × 2πr = (60/360) × 2 × (22/7) × 21 = (1/6) × 132 = **22 cm**

(ii) Area of sector = (60/360) × (22/7) × 21 × 21 = (1/6) × 22 × 63 = **231 cm²**

(iii) OA = OB = 21 cm and ∠AOB = 60°, so ∠OAB = ∠OBA = 60°: ΔOAB is equilateral with side 21 cm.

Area of ΔOAB = (√3/4) × 21² = 441√3/4 cm²

Area of segment = sector − triangle = **(231 − 441√3/4) cm²** (≈ 40.05 cm²)

6 **A chord of a circle of radius 15 cm subtends an angle of 60° at the centre. Find the areas of the corresponding minor and major segments of the circle. (Use π = 3.14 and √3 = 1.73)**

✔ ANSWER

r = 15 cm, θ = 60°

Area of minor sector = (60/360) × 3.14 × 15 × 15 = 706.5/6 = 117.75 cm²

ΔOAB is equilateral (OA = OB, ∠AOB = 60°), side 15 cm

Area of ΔOAB = (√3/4) × 15² = (1.73/4) × 225 = 97.3125 cm²

Area of minor segment = 117.75 − 97.3125 = **20.4375 cm²**

Area of circle = 3.14 × 15² = 706.5 cm²

Area of major segment = 706.5 − 20.4375 = **686.0625 cm²**

7 **A chord of a circle of radius 12 cm subtends an angle of 120° at the centre. Find the area of the corresponding segment of the circle. (Use π = 3.14 and √3 = 1.73)**

✔ ANSWER

r = 12 cm, θ = 120°

Area of sector OAPB = (120/360) × 3.14 × 12 × 12 = 452.16/3 = 150.72 cm²

Draw OM ⊥ AB. Then ∠AOM = 60°, AM = MB.

OM = OA cos 60° = 12 × ½ = 6 cm

AM = OA sin 60° = 12 × (√3/2) = 6√3 cm, so AB = 12√3 = 12 × 1.73 = 20.76 cm

Area of ΔOAB = ½ × AB × OM = ½ × 20.76 × 6 = 62.28 cm²

Area of segment = 150.72 − 62.28 = **88.44 cm²**

8 **A horse is tied to a peg at one corner of a square shaped grass field of side 15 m by means of a 5 m long rope (see Fig. 11.8). Find (i) the area of that part of the field in which the horse can graze. (ii) the increase in the grazing area if the rope were 10 m long instead of 5 m. (Use π = 3.14)**

✔ ANSWER

The angle at a corner of the square is 90°, so the horse grazes a quadrant (sector of 90°) of radius = length of rope.

(i) r = 5 m: Area = (90/360) × 3.14 × 5 × 5 = ¼ × 78.5 = **19.625 m²**

(ii) r = 10 m: Area = ¼ × 3.14 × 10 × 10 = 78.5 m²

Increase in grazing area = 78.5 − 19.625 = **58.875 m²**

9 **A brooch is made with silver wire in the form of a circle with diameter 35 mm. The wire is also used in making 5 diameters which divide the circle into 10 equal sectors as shown in Fig. 11.9. Find: (i) the total length of the silver wire required. (ii) the area of each sector of the brooch.**

✔ ANSWER

Diameter = 35 mm, so r = 35/2 mm

(i) Circumference = 2πr = 2 × (22/7) × (35/2) = 110 mm

Length of 5 diameters = 5 × 35 = 175 mm

Total wire = 110 + 175 = **285 mm**

(ii) 10 equal sectors, so angle of each = 360°/10 = 36°

Area of each sector = (36/360) × (22/7) × (35/2) × (35/2) = (1/10) × (22/7) × 1225/4 = 3850/40

Area of each sector = 385/4 mm² (= 96.25 mm²)

10 **An umbrella has 8 ribs which are equally spaced (see Fig. 11.10). Assuming umbrella to be a flat circle of radius 45 cm, find the area between the two consecutive ribs of the umbrella.**

✔ ANSWER

8 equally spaced ribs make 8 equal sectors; angle of each = 360°/8 = 45°

Area between two consecutive ribs = (45/360) × (22/7) × 45 × 45

= (1/8) × (22/7) × 2025 = 44550/56

Area = 22275/28 cm² (≈ 795.54 cm²)

11 **A car has two wipers which do not overlap. Each wiper has a blade of length 25 cm sweeping through an angle of 115°. Find the total area cleaned at each sweep of the blades.**

✔ ANSWER

Each wiper cleans a sector with r = 25 cm, θ = 115°

Area cleaned by one wiper = (115/360) × (22/7) × 25 × 25 = (23/72) × (13750/7) = 158125/252 cm²

Total area (two wipers) = 2 × 158125/252

Total area cleaned = 158125/126 cm² (≈ 1254.96 cm²)

12 **To warn ships for underwater rocks, a lighthouse spreads a red coloured light over a sector of angle 80° to a distance of 16.5 km. Find the area of the sea over which the ships are warned. (Use π = 3.14)**

✔ ANSWER

r = 16.5 km, θ = 80°

Area = (80/360) × 3.14 × 16.5 × 16.5

= (2/9) × 3.14 × 272.25 = (2/9) × 854.865

= 1709.73/9

Area ≈ 189.97 km²

13 **A round table cover has six equal designs as shown in Fig. 11.11. If the radius of the cover is 28 cm, find the cost of making the designs at the rate of ₹ 0.35 per cm². (Use √3 = 1.7)**

✔ ANSWER

Six equal designs: each design is a segment made by a chord, with angle at the centre = 360°/6 = 60°.

r = 28 cm. Area of one sector = (60/360) × (22/7) × 28 × 28 = (1/6) × 2464 = 410.67 cm² (approx.)

The triangle with the chord is equilateral (OA = OB, ∠AOB = 60°), side 28 cm

Area of triangle = $(\sqrt{3}/4) \times 28^2 = (1.7/4) \times 784 = 333.2$ cm²

Area of one design (segment) = 410.67 − 333.2 = 77.47 cm² (approx.)

Area of six designs = 6 × (2464/6 − 333.2) = 2464 − 1999.2 = 464.8 cm²

Cost = 464.8 × 0.35 = **₹ 162.68**

14 **Tick the correct answer in the following: Area of a sector of angle p (in degrees) of a circle with radius R is**

(A) $(p/180) \times 2\pi R$ (B) $(p/180) \times \pi R^2$ (C) $(p/360) \times 2\pi R$ (D) $(p/720) \times 2\pi R^2$

✔ ANSWER

(D) $(p/720) \times 2\pi R^2$

Area of sector = $(p/360) \times \pi R^2$. Writing $(p/360) \times \pi R^2 = (p/720) \times 2\pi R^2$ gives option (D).

(A) and (C) have $2\pi R$, which is a length (arc), not an area; (B) is twice the correct value.

🎉 **Chapter 11 done! Revise the answers once more tomorrow.**